

Math 290

Practice Quiz 2 — Version D Solutions

1(a) Answer: C. The theorem is true, but the proof splits into the cases “ x even” and “ y even,” which is assuming the conclusion; nothing rules out x and y both odd. (See Warning 8.7: to *prove* $P \vee Q$, the cases should be P and $\neg P$.) A correct proof, contrapositively: if $x = 2k + 1$ and $y = 2\ell + 1$, then $xy = 4k\ell + 2k + 2\ell + 1 = 2(2k\ell + k + \ell) + 1$ is odd. (Based on Proposition 8.5 and Warning 8.7.)

1(b) Answer: False. If $x < 0$, then $-x > 0$, so $|-x| = -x$, which is positive and hence $\neq x$. For example $x = -3$: $| -(-3) | = 3 \neq -3$. (Based on Warning 8.19.)

1(c) Answer: True. $\neg(\forall x, P(x) \Rightarrow Q(x)) \equiv \exists x, P(x) \wedge \neg Q(x)$. (Based on Sections 5 and 7.C.)

2. (Based on Exercise 7.7.)

Solution. (a) Let $a \in \mathbb{Z}$. We work contrapositively. Assume a is even, so $a = 2k$ for some $k \in \mathbb{Z}$. Then $a^2 = 4k^2$, and since $k^2 \in \mathbb{Z}$, we have $4 \mid a^2$.

(b) The converse is: “If a is odd, then $4 \nmid a^2$.” This is *true*. Proof by contradiction: assume a is odd and $4 \mid a^2$. Write $a = 2k + 1$ and $a^2 = 4m$ with $k, m \in \mathbb{Z}$. Then $4m = 4k^2 + 4k + 1$, so $1 = 4(m - k^2 - k)$, meaning $4 \mid 1$, which is false. Hence $4 \nmid a^2$. $\square$

3. (Based on Exercise 9.5.)

Solution. Lemma: if $a \in \mathbb{Z}$ and a^3 is even, then a is even. Contrapositively, if $a = 2k + 1$, then $a^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1$ is odd.

Assume, by way of contradiction, that $\sqrt[3]{2} \in \mathbb{Q}$. Write $\sqrt[3]{2} = a/b$ with $a, b \in \mathbb{N}$ in lowest terms. Cubing and clearing denominators gives $a^3 = 2b^3$, so a^3 is even and, by the lemma, a is even. Write $a = 2m$ with $m \in \mathbb{Z}$. Then $8m^3 = 2b^3$, so $b^3 = 4m^3 = 2(2m^3)$ is even, and by the lemma b is even. Thus a and b are both even, contradicting lowest terms. Hence $\sqrt[3]{2}$ is irrational. $\square$

4. (Based on Exercise 10.2.)

Solution. ($A \subseteq B$) Let $(x, y) \in A$. Then $x, y \in \mathbb{Z}$ and $4 \mid (x - y)$, so $x - y = 4k$ for some $k \in \mathbb{Z}$, i.e. $y = x - 4k$. Case 1: x is even, say $x = 2m$. Then $y = 2m - 4k = 2(m - 2k)$ is even. Case 2: x is odd, say $x = 2m + 1$. Then $y = 2m + 1 - 4k = 2(m - 2k) + 1$ is odd. In both cases x and y have the same parity, and $(x, y) \in \mathbb{Z} \times \mathbb{Z}$, so $(x, y) \in B$.

($B \subseteq A$ is false.) Counterexample: $(0, 2)$. Both entries are even, so $(0, 2) \in B$, but $0 - 2 = -2$ and $4 \nmid -2$, so $(0, 2) \notin A$. $\square$